

Tianyu Chen

☎ (+1) 773-431-6690

✉ tianyuchen@utexas.edu

🔗 Google Scholar

🔗 TianyuCodings

EDUCATION

University of Texas at Austin	Ph.D. in Statistics (GPA: 4.0), Austin, TX	Oct. 2023 – July 2028 (expected)
University of Chicago	M.S. in Statistics (GPA: 3.97), Chicago, IL	Oct. 2021 – July 2023
Fudan University	B.S. in Statistics (GPA: 3.7), Shanghai, China	Sept. 2017 – July 2021

INDUSTRY EXPERIENCE

Microsoft Research | MAI Superintelligence

Research Scientist Intern (Mentors: Zhendong Wang, Lijuan Wang)

Feb. 2025 – March. 2026

- Developed an object-centric vision-language model (VLM) agent that automates fine-grained, multi-turn image-editing evaluation, eliminating reliance on human annotators for benchmark construction. The project received media coverage in well-known Chinese outlets and is showcased on the project website.
- Led large language model (LLM) post-training research on mathematical reasoning, LLM-as-a-judge alignment, and on-policy-distillation, including recipe design for **autoregressive** and **diffusion-based** language models.

Google Deepmind | Gemini Embedding

Research Scientist Intern (Mentors: Jiaxing Wu)

March. 2025 – Present

- Gemini Embedding model's post training. I am designing the RL algorithm to enhance the embedding model's instruction following ability on compositional image retrieval tasks.

MORE INFORMATION

LinkedIn: <https://www.linkedin.com/in/tianyu-chen-1a056a160/>

Website: <https://tianyucodings.github.io/>

Research Interests: LLM post-training (RL and on-policy-distillation), image editing, and model distillation

GENERATIVE MODEL & REINFORCEMENT LEARNING

- [Preprint] **Tianyu Chen***, Yasi Zhang*, Mingyuan Zhou, Oscar Leong, Ying Nian Wu, Michal Lukasik[†]. “REAL: Regression-Aware Reinforcement Learning for LLM-as-a-Judge.” (*Collaboration with Google DeepMind*).
- [Neurips 2024] **Tianyu Chen**, Zhendong Wang, Mingyuan Zhou. “Diffusion Policies creating a Trust Region for Offline Reinforcement Learning.” Published in: *Neurips 2024*. [arXiv]. 2024.
- [ICLR 2026] **Tianyu Chen***, Yasi Zhang*, Zhi Zhang, Peiyu Yu, Shu Wang, Zhendong Wang, Kevin Lin, Xiaofei Wang, Zhengyuan Yang, Linjie Li, Chung-Ching Lin, Jianwen Xie, Oscar Leong, Lijuan Wang, Ying Nian Wu, Mingyuan Zhou. “EdiVal-Agent: An Object-Centric Framework for Automated, Scalable, Fine-Grained Evaluation of Multi-Turn Editing.” Accepted to: *ICLR 2026*. [OpenReview]. 2026.
- [ICLR 2026] **Tianyu Chen***, Yasi Zhang*, Zhendong Wang, Ying Nian Wu, Mingyuan Zhou. “Score Distillation Beyond Acceleration: Generative Modeling from Corrupted Data.” Accepted to: *ICLR 2026*. [OpenReview]. 2026.
- [Preprint] **Tianyu Chen***, Xinran Song*, Mingyuan Zhou. “A Generative Framework for Causal Estimation via Importance-Weighted Diffusion Distillation.” Preprint [arXiv]. 2025.
- [Neurips 2025] Yifan Sun, Jingyan Shen, Yibin Wang, **Tianyu Chen**, Zhendong Wang, Mingyuan Zhou, Huan Zhang. “Improving Data Efficiency for LLM Reinforcement Fine-tuning Through Difficulty-targeted Online Data Selection and Rollout Replay.” Accepted to: *Neurips 2025*. [arXiv]. 2025.
- [PNAS] **Tianyu Chen***, Jin-Hong Du*, Ming Gao, Jingshu Wang. “Model-based trajectory inference for single-cell rna sequencing using deep learning with a mixture prior.” Published in: *PNAS*. [PNAS]. 2024.

STATISTICS

- [Neurips 2025 Spotlight] **Tianyu Chen**, Vansh Bansal, James G. Scott. “CoLT: The conditional localization test for assessing the accuracy of neural posterior estimates.” **Spotlight at: Neurips 2025**. [arXiv]. 2025.
- [ICML 2026 Submission] **Tianyu Chen**, Vansh Bansal, James G. Scott. “The surprising strength of weak classifiers for validating neural posterior estimates.” Submitted to: *ICML 2026*. [arXiv]. 2025.
- [AISTATS 2025] **Tianyu Chen**, James G. Scott. “Conditional diffusions for neural posterior estimation.” [arXiv]. 2025.
- [Neurips 2024] **Tianyu Chen**, Kevin Bello, Francesco Locatello, Bryon Aragam, Pradeep Ravikumar. “Identifying General Mechanism Shifts in Linear Causal Representations.” Published in: *Neurips 2024*. [arXiv]. 2024.
- [Neurips 2023] **Tianyu Chen**, Kevin Bello, Bryon Aragam, Pradeep Ravikumar. “iSCAN: Identifying Causal Mechanism Shifts among Nonlinear Additive Noise Models.” Published in: *Neurips 2023*. [arXiv]. 2023.